

ARUN KONAPALA

Lead AI / GenAI Platform Engineer · AWS Bedrock · Agentic AI · LangChain · Python
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SUMMARY

Lead Software Engineer with 20 years of enterprise and AI engineering experience, currently at Fannie Mae architecting agentic AI platforms, LLM evaluation frameworks, and MCP-integrated GenAI systems on AWS. Deep expertise in AWS Bedrock, LangChain, RAG pipelines, multi-agent orchestration (MCP, AWS AgentCore, LangGraph), and Python microservices across financial services and mortgage data domains. Proven record of delivering enterprise-wide AI platforms — from observability frameworks and text-to-SQL agents to Angular-based AI ops dashboards — adopted across business units and meeting regulatory compliance requirements.

EXPERIENCE

Fannie Mae

Lead Software Engineer · Contract

Reston, VA

Apr 2016 – Present

- Led a team of 10 engineers in design discussions with Enterprise Architects, Infosec Architects, and Vendors — integrating MCP servers with MCP Gateway and APIGEE to enable standardized AI tool connectivity across the enterprise.
- Collaborated with Enterprise Architects and Infosec Architects to enable AWS Bedrock AgentCore Runtime, AgentCore Memory, and AgentCore Code Interpreter — supporting business use cases requiring dynamic analytics over loan forecast data.
- Designed multi-layer observability framework for AI inference pipeline with real-time metrics enrichment, KMS encryption, and audit compliance — standardized observability across 3 Lambda workloads enabling cost attribution and model health monitoring at scale.
- Designed and implemented queue-based orchestrator with SQS poller → RawIngestionStage (buffer/flush) → AggregationStage cascade (1min→5min→hourly→daily→monthly), using DirtyPartitionTracker for late-arriving data recovery and CDHBARScheduler for daily metadata enrichment.
- Designed and implemented FastAPI monitoring & management platform for AI/ML operations — tracks utilization and performance with RBAC auth, camelCase JSON responses, and dynamic role binding to AD groups.
- Designed and implemented standalone Angular SPA with signals-based state management, lazy-loaded feature routes, and strict TypeScript — featuring AI/ML ops dashboards covering Metrics, Models, Traces & Logs, Agents Registry, Budget Tracking, Cost Optimization, and Onboarding flows, serving 1,000+ users across business units.
- Reduced model quality risk enterprise-wide by designing and implementing a robust LLM Evaluation Framework that benchmarks 40+ foundational models and GenAI applications — enabling continuous monitoring and early detection of performance drift.
- Standardized enterprise data ingestion by architecting and implementing a configuration-driven Hydration Framework adopted org-wide — eliminating bespoke pipelines and reducing onboarding time for new data sources from weeks to days.
- Designed and implemented the Operational Insights Hub (now Lineage Explorer) — the single source of metadata, controls, STTM documents, and operational metrics for enterprise users; leveraged Neo4j graph database for end-to-end lineage visualization meeting regulatory expectations.
- Designed and implemented an enterprise application for Model Risk Management reporting on Model Usage Inventory System (MUSE) data using AWS — addressing the reporting needs of 1LOD, 2LOD, 3LOD, and regulators.

Various Clients (Independence Blue Cross, Blue Cross Blue Shield Association, Societe Generale CIB)

IL / PA

Liferay Portal Architect · Contract

Apr 2014 – Mar 2016

- Architected and delivered Liferay portal solutions across healthcare and financial services clients — including the BlueWeb member portal (BCBS), IBX sales portal, and a self-service trading module for SG CIB — improving usability, scalability, and role-based access across all platforms.
- Implemented Apache SOLR-based full-text search across multiple portals — reducing query latency and improving content discoverability for members, sales representatives, and administrators.
- Ensured high availability by configuring multi-node Liferay clusters and leading post-migration performance tuning — meeting production SLAs and minimizing downtime risk across client environments.

Mahindra Satyam

Liferay Team Lead · Full-time

United States

Jul 2010 – Mar 2014

- Led a team of engineers across multiple client engagements by providing technical direction on Liferay portal architecture, RESTful service design, and Java development — consistently delivering projects on time across GE Transportation, Cisco, Computer Associates, and Thomson Reuters.
- Enabled system interoperability across customer-facing portals by designing and implementing RESTful and SOAP-based web services — providing standardized data exchange interfaces for downstream consumers and third-party integrations.
- Improved application responsiveness across multiple client portals by profiling and performance-tuning Spring and Hibernate modules — resolving bottlenecks and delivering faster, more reliable user experiences.
- Secured enterprise access by integrating Ping Identity and SSO systems for authentication and authorization — enforcing role-based controls and meeting enterprise security standards across client environments.

Prithvi Information Solutions Ltd

Senior Software Engineer · Full-time

United States

Jun 2006 – Jul 2010

- Delivered Java-based enterprise applications across client engagements — building reusable modules, data validation layers, and Identity & Access Management integrations that enforced role-based governance and compliance across organizations.

PROJECTS

Source code available at github.com/arunkonapala

Finance Copilot: Conversational Personal Finance Assistant

Python · FastAPI · Claude · Tool Use · SSE · Angular

- Built a conversational finance copilot with a streaming agentic tool-use loop on Anthropic Claude — eight finance tools spanning spending analysis, budgets, bills, and portfolio Q&A, streamed token-by-token over SSE to an Angular 20 chat UI with a provider-pluggable backend supporting OpenAI-compatible endpoints.

FinDocQA: Financial Document Q&A with RAG

Python · LangChain · FAISS · Cohere · Claude · AWS S3 · Angular

- Built a RAG pipeline over annual reports and SEC filings using FAISS vector search with Cohere embeddings and LangChain — delivering grounded Claude answers with page-level citations through an Angular UI, with document storage on AWS S3.

AI Nutrition Coach: Meal Photo Analysis

Python · FastAPI · Claude Vision · Multimodal AI · Angular

- Built a multimodal nutrition assistant that analyzes meal photos with Claude vision agents — returning structured nutrient breakdowns and recipe ideas via a FastAPI backend and an Angular 20 interface.

AutoMed: Multi-Agent Healthcare Chatbot

Python · AG2 (AutoGen) · Multi-Agent AI · LLMs

- Built a multi-agent healthcare chatbot using AG2 (AutoGen) orchestrating specialized agents for symptom analysis, treatment suggestions, and follow-up care — delivering health-history-aware, context-sensitive recommendations that simulate expert medical team collaboration.

Meal & Grocery Planner with CrewAI

Python · CrewAI · Pydantic · LLMs · YAML · Agentic AI

- Built an end-to-end meal planning system using CrewAI agents for recipe research, shopping organization, and budget optimization — backed by a CrewBase architecture with Pydantic models and declarative YAML config for a reusable, maintainable agentic pipeline.

Style Finder: Computer Vision Fashion Analysis

Python · Llama 4 · Multimodal RAG · Computer Vision · Angular · Vector Search

- Built a multimodal RAG pipeline using Llama 4 Maverick 17B and image encoding for vector-based fashion similarity search — enabling users to upload photos and receive catalog-style AI insights on garments and styling elements via an Angular web application.

DataWizard: AI-Powered Data Analysis Agent

Python · LangChain · LLMs · Scikit-learn · Agentic AI

- Built a LangChain-powered AI agent enabling non-technical users to perform CSV analysis, statistical summarization, and ML model training through natural language — using a multi-step executor agent to orchestrate complex analytical workflows.

AI Meeting Assistant

Python · Whisper · AWS Bedrock · LangChain · React · LLMs

- Built an AI meeting assistant integrating Whisper STT with AWS Bedrock and LangChain PromptTemplate workflows — automating capture of decisions, action items, and deadlines from audio recordings into structured meeting minutes via a React interface.

Interactive Food Search & RAG Chatbot System

Python · Chroma DB · RAG · LLMs · NLP · CLI

- Built a food recommendation system combining Chroma DB vector search and RAG with advanced metadata filtering (cuisine, calories, ingredients) and CRUD operations — delivering AI-enhanced conversational recommendations across three distinct search approaches.

SKILLS

Generative AI: AWS Bedrock, AWS AgentCore, GitHub Copilot, Claude, ChatGPT, Hugging Face, Guardrails AI, Portkey, MCP (Model Context Protocol)

AI / ML: SageMaker, DeepEval, LangChain, LangGraph, LangSmith, LlamaIndex, CrewAI, AutoGen (AG2), Strands Agents, Whisper, Scikit-learn, NumPy, Pandas, Matplotlib, PyTorch, TensorFlow

Backend: Spring Boot, Flask, FastAPI, Hibernate, Pydantic

Web Services: RESTful APIs, GraphQL

Compute & Containers: Elastic Beanstalk, EC2, ECS, AWS Lambda, EMR, Kubernetes, Apache Spark

Messaging & Events: SNS, SQS, EventBridge, Apache Kafka

Security & Compliance: SonarQube, Ping Identity SSO, Secrets Manager, IAM, KMS

Infrastructure as Code: CloudFormation, Terraform

CI/CD & Deployment: Jenkins, GitLab, GitHub Actions, CodePipeline, Docker

AI Techniques: RAG, Prompt Engineering, Fine-tuning, Vector Embeddings, NLP, Computer Vision, Multimodal AI

Languages: Python, Java, TypeScript

Frontend: Angular, React, Next.js, Streamlit

Data Stores: S3, PostgreSQL, MySQL, DynamoDB, Redshift, Neo4j, Oracle, FAISS, Chroma DB, pgvector, Redis, Elasticsearch

Network & Delivery: API Gateway, Apigee, Route 53, VPC, Elastic Load Balancing

Monitoring & Logging: CloudWatch, CloudTrail, X-Ray, Dynatrace, Splunk, OpenTelemetry, Google Analytics 4

Workflow & Orchestration: AWS Step Functions, AWS Glue

Build Tools: Maven, npm

CERTIFICATIONS

IBM — 10 AI & GenAI Certifications (2025)

Agentic AI with LangGraph, CrewAI, AutoGen & BeeAI · IBM RAG and Agentic AI · Multimodal GenAI Applications · Advanced RAG with Vector Databases · +6 GenAI & Agentic AI courses

Databricks (2025)

Academy Accreditation – Generative AI Fundamentals

Amazon Web Services (2025)

Generative AI for Developers · GenAI Applications with Amazon Bedrock · Getting Started with AWS GenAI

Pegasystems & Sun Microsystems

PRPC CSA – Pegasystems · SCJP – Sun Microsystems

EDUCATION

SRKR Engineering College

Bachelor of Engineering

India
Aug 2000 – Oct 2004